# Data_Scientist — Step 5

## Build Data Science Projects

Detailed Learning Notes • CareerCompass

## 1. Learning Objective

Combine data acquisition, statistics, analysis, modeling and communication into complete portfolio-quality projects.

## 2. Core Concepts — Detailed Breakdown

### End-to-end structure

A strong project moves from problem definition to data, EDA, baseline, modeling, evaluation, interpretation and recommendation.

### Data acquisition

Use APIs, databases, CSVs or public datasets. Document source, access date, licensing and known limitations.

### Experiment design

Create a baseline and maintain a consistent evaluation protocol. Record experiments rather than relying on memory.

### Communication

Present technical findings differently to technical and non-technical audiences.

### Reproducibility

Include requirements, setup instructions, data preparation steps, notebooks/scripts and clear output.

## 3. Recommended Learning Workflow

- Select a question with measurable value.
- Create a project plan and data dictionary.
- Build an analysis baseline.
- Add a predictive or inferential component.
- Document limitations.
- Publish a clean repository and report.

## 5. Hands-on Project / Practice

Build a demand-forecasting, churn-analysis or customer-segmentation project. Include raw-data acquisition, cleaning, EDA, a statistical or ML component, evaluation, visualization, recommendations and a README explaining how another person can reproduce it.

## 6. Common Mistakes & Pitfalls

- Building a tutorial clone without independent decisions.
- No clear question or success criterion.
- Mixing exploratory and final conclusions without labeling them.

- Hiding failed experiments.
- Publishing sensitive or licensed data improperly.

## 7. Completion Checklist

- I have a clear research/business question.
- My data source and assumptions are documented.
- My analysis is reproducible.
- My model/evidence is evaluated appropriately.
- My final report contains actionable conclusions and limitations.

## 8. Interview / Assessment Questions

- Explain your project in two minutes.
- What failed and what did you learn?
- How did you validate the result?
- What would you do with more data?
- How would you communicate this to a non-technical manager?

## 9. Recommended References

Data Science Tutorial: https://www.youtube.com/watch?v=ua-CiDNNj30

Python for Data Science: https://www.youtube.com/watch?v=CMEWVn1uZpQ

Kaggle Learn: https://www.kaggle.com/learn

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.